PROVEIT RESEARCH NOTE

Signed growth and centered Pell transport

Explicit Diophantine equations for the consistency of ZFC

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25 October 2026

Abstract

We give explicitly specified ordinary polynomials whose nonsolvability, in their stated domains, is provably equivalent in ZFC to syntactic Con(ZFC). A signed affine horizon, followed by a divisibility-justified rescaling, yields a quartic in 209 unrestricted integers. A separate construction combines a free quotient with an integer polynomial center and a new transport bound to obtain degree 190 in ten positive integers. Three compositions with oriented pooling give further unrestricted-integer tradeoffs. Every equation is accompanied by its literal coefficient or circuit data, constructive reductions, fixed-source provenance, and reproducible exact checks. We also identify why a tempting additional variable elimination fails: it limits the capacity of the proof-state encoding. The results improve specified constructions and do not establish a global minimum for Diophantine consistency equations.

Proof status. The new arguments are ordinary mathematical proofs formalizable in ZFC, composed with the explicitly named inherited refutation calibrations. Exact computation authenticates finite polynomial identities and numerals. A new end-to-end Lean/Rocq proof object is not claimed. Mathematical development and independent internal reviews were assisted by ChatGPT; they are not external human peer review.

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1 Results, meaning, and comparison

The central unrestricted-integer deliverable is a fully specified polynomial

$$Q_{209}(\xi_0, \dots, \xi_{208}) = \sum_{j=0}^{148} R_j(\boldsymbol{\xi})^2. \tag{1}$$

Every R_j is an explicitly listed polynomial in $\mathbb{Z}[\boldsymbol{\xi}]$ of degree at most two. Appendix A prints every residual; `quartic/equations/affine209_F64/equation.json` and its collected expansion are the corresponding exact data. No parameter remains to be chosen by an undecidable procedure.

Theorem 1.1 (Consistency equivalence)**.** *For the fixed polynomial Q_{209} in (1),*

$$\mathsf{ZFC} \vdash \Big(\neg\exists \boldsymbol{\xi} \in \mathbb{Z}^{209}\ Q_{209}(\boldsymbol{\xi}) = 0\Big) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The same statement holds for the delivered second quartic and the three pooled polynomials, in their respective unrestricted-integer dimensions. The separately specified polynomial P_{10} of Section 3 satisfies

$$\mathsf{ZFC} \vdash \Big(\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{10}\ P_{10}(\boldsymbol{x}) = 0\Big) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The proof composes the new arithmetic reductions below with the precise inherited calibrations in Section 6. The consistency predicate is the absence of a finite syntactic refutation in the repository's fixed effective presentation `ZFCax_s`, with proof relation `BProv` and contradiction `Form.fBot`. The quantifiers are internal to ZFC; the assertion does not restrict proof codes to externally standard integers. It assumes neither consistency nor inconsistency and proves neither one.

1.1 The delivered profiles

Profile	Variables	Degree	Squares	Monomials	Height / cost
Sparse quartic, $\mathbb{Z}$	209	4	149	5,486	height 2^{32}
Lower-height quartic, $\mathbb{Z}$	209	4	149	6,654	height 2^{30}
Pooled, $\mathbb{Z}$	130	12	70	circuit	2,475 operations
Pooled, $\mathbb{Z}$	116	16	56	circuit	2,433 operations
Pooled, $\mathbb{Z}$	107	32	47	circuit	2,419 operations
Centered transport, $\mathbb{Z}_{>0}$	10	190	–	circuit	119 operations

For the quartics, height is the largest absolute coefficient of the *collected final polynomial*, not a literal in its evaluator. All monomial counts also refer to collected final polynomials. For the circuit profiles, the finite straight-line program is the complete polynomial specification; a full expanded monomial count is not claimed. An operation is one live binary addition, subtraction, or multiplication; squaring and multiplication by a fixed integer are charged, while reuse is free. This is an arithmetic count, not a bit-complexity estimate.

The inspected repository snapshot is

`d123610bc98fef2006bff1c9466bc3f1221003e7`.

It contains placed packages through `178-polynomial-centers`, together with seven incoming ZIP archives. Comparisons below include those incoming archives rather than treating the last placed package as the only baseline.

The immediate unrestricted quartic parent has 210 variables, 150 squared quadratics, 5,282 monomials, and height 2^{40}. The sparse new quartic saves one variable and one square and divides the height by 256, at the cost of 204 additional monomials. The second new quartic divides the parent's height by 1,024, at a larger support cost. These are explicit tradeoffs, not simultaneous improvements in every measure.

The corresponding older pooled profiles have 131 variables at degree 12, 117 variables at degree 16, and 108 variables at degree 32. The new composition saves one variable in each case, with unchanged degrees. For ten positive variables the relevant points are source 172's degree 214 at 87 operations, source 178's degree 208 at 116 operations, and the incoming affine-centering result's degree 206 at 90 operations. Our degree 190 costs 119 operations. It reduces the lowest degree among these inspected ten-positive-variable constructions by 16, approximately 7.77%, while using more operations.

There is no claim that 209 is the least possible integer dimension, that 190 is the least possible degree with ten positive variables, or that these constructions are globally simplest. The repository already contains much smaller-variable representations in other degree and coefficient regimes. The word "simplest" requires a choice among dimension, degree, coefficient height, support, evaluation length, and proof complexity.

1.2 What is new in this iteration

The first reduction replaces a separate positive square horizon by $t = 2 + 4z$, and recovers its sign from the remaining exponent and endpoint certificates. A large *absolute* lower bound suffices before the sign is known. This allows the sharp positive-center Pell bound to be applied only after $t > 0$, avoiding an incorrect estimate on negative centers. The retained divisibility then permits an integral rescaling of the growth coordinate. A change of residual basis yields a second height tradeoff.

The compact reduction uses an integer polynomial approximation to an entire Pell expression. The free quotient in its source admits a small auxiliary index that invalidates a naive transport estimate. The new transport $L = UEb^2$ gives a uniform bound including that index, while keeping the cross term below the degree of the centered norm defect.

The pooled reductions compose the corrected signed growth interface with four-base reciprocal pooling and explicit acyclic eliminations. The pool does not need pairwise coprime moduli. Its soundness comes from a strict bound on a signed rational sum of squared errors.

1.3 Why a different formal calculus alone is insufficient

The direct branch uses the numerical calculus $\mathcal{C}_{222}$ through its independent refutation calibration; it does not simulate the K215 machine or instantiate Jones's universal polynomial. The compact branch retains its native/compiler/K215 ancestry. These are different constructions.

Alternative logical encodings remain relevant, but they need a complete unbounded compiler and explicit coefficients before yielding a competitive equation. Formisano, Gavazzi, and Omodeo explore a relation-algebra presentation of NBG in this direction [9]. General variable-count theorems, including Sun's recent ten-integer-variable representation, likewise do not by themselves specify a particularly small fixed consistency instance [7]. The present report measures actual emitted polynomials.

2 A signed horizon and an integral rescaling

2.1 The exact arithmetic interface

We retain the 210-variable parent's literal signed Pell, binomial, and Chinese remainder construction [2]. Its horizon, originally $t = 16 + 4z^2$, is changed to

$$t = 2 + 4z, \qquad z \in \mathbb{Z}. \tag{2}$$

Write $d = 2^{19}$. The three standalone growth equations remain

$$Z = tF, \qquad X^2 - 16Z^2 - 8ZF = 1, \qquad F = dt(2u + 1). \tag{3}$$

At this stage t, F, X, Z, u are signed. Every displayed equation has degree at most two. The Pell equation obtained by substituting the first row is $X^2 - ((4t + 1)^2 - 1)F^2 = 1$; this substituted expression is for the proof, not an additional higher-degree residual in the quartic.

The parent also retains an even anchor C, a binomial selector B, and the identities

$$B\eta = C + F + 1, \qquad D = 2(C + F + 1). \tag{4}$$

Its ordinary polynomial power certificates have the semantic consequences

$$q = D^x, \qquad q^t, \qquad q^{y+1}, \qquad (q + 1)^{F+t}, \qquad (q + 1)^C, \tag{5}$$

as well as the other requested powers in the exact source. A notation such as q^t in this interface means that the retained polynomial certificate has a proposed integral output for this base and exponent. It is not a variable-exponent expression in the final polynomial. For $|q| \geq 2$, the certificate forces its exponent to be nonnegative and its output to be the true power.

The beta endpoints are

$$\begin{aligned} y &= F(2v + 1), \\ x &= F + 1 + (y + 1)r_0, \\ x &= 1 + (1 + (t + 1)y)r_t. \end{aligned} \tag{6}$$

The letters in this interface are semantic names for fixed source coordinates or polynomial abbreviations. The final variable list is given independently in the literal equation data.

2.2 Recover the signs before applying a sharp growth bound

Lemma 2.1 (Preliminary bound)**.** *Every integer solution of* (2) *and the last row of* (3) *satisfies*

$$t \neq 0, -1, \quad |t| \geq 2, \quad F \neq 0, \quad 2^{20} \mid F, \quad |F| \geq d|t| > |t|.$$

Proof. The progression in (2) is exactly the integers congruent to 2 modulo 4. Thus $\nu_2(t) = 1$. The integer $2u + 1$ is nonzero and odd, so $\nu_2(F) = 20$, and its absolute value is at least one. □

The parent's square frame includes the actual growth coordinate F. Consequently its parameter W is at least $2 + F^2 > 258$, even though t^2 is now permitted to be smaller than 256. This supplies the original common-frame threshold before signs are recovered. The cap and orientation proof uses squares and absolute values: it classifies the anchor and all principal Pell pairs at their canonical signed indices without assuming $t > 0$ or $F > 0$. The new standalone pair in (3) is independent of this common frame and needs no new frame atom. All actual bases, nominal indices, and outputs retained from the parent remain in the frame.

Lemma 2.2 (Noncircular sign recovery)**.** *With the retained canonical-index and exact-power interface, every zero of the changed source has*

$$t \geq 2, \quad x \geq 2, \quad y \geq F > 0, \quad C > 0, \quad D > 2C > 0, \quad q > 0.$$

Proof. Both C and F are even. Thus $C + F + 1$ is odd and nonzero; (4) makes B, η odd and nonzero, and $|D| \geq 2$. The request $q = D^x$ gives $x \geq 0$.

Set $g = 1 + (t+1)y$. By Lemma 2.1 and (6),

$$|g| \geq |t+1||y| - 1 \geq |F| - 1 > 1,$$

because $t \neq -1$. If $x = 0$, the last endpoint would give $gr_t = -1$, a contradiction. Therefore $x \geq 1$, and $|q| \geq 2$. The retained request with exponent t, exactly t rather than $t+1$, now forces $t \geq 0$. The progression excludes zero, so $t \geq 2$.

The request with exponent $y + 1$ gives $y + 1 \geq 0$. A negative y would have $y \leq -|F| < -1$. Hence $y \geq |F| > 0$. If $x = 1$, the first endpoint says $(y+1)r_0 = -F$, impossible since $0 < |F| \leq y < y + 1$. Thus $x \geq 2$, $|q| \geq 4$, and $|q + 1| \geq 3$.

The requested $(q+1)$-powers now imply $F + t \geq 0$, $C \geq 0$, and nonnegativity of every other actual CRT index. A negative F would give $F + t = t - |F| < 0$, so $F > 0$. The anchor proof already excludes $C = 0$, yielding $C > 0$. Formula (4) gives $D > 2C > 0$ and then $q = D^x > 0$. □

This order matters. On a negative branch the positive Pell center would be $|4t + 1| = 4|t| - 1$. One cannot apply a lower bound proved for $4|t| + 1$ there. The elementary inequality $|F| \geq d|t|$ supplies all the initial sign information we need.

2.3 The sharp positive-center estimate works at the smallest horizon

For $a \geq 2$, write $X_n(a), Y_n(a)$ for the ordinary Pell sequences

$$X_n(a) + Y_n(a)\sqrt{a^2 - 1} = \left(a + \sqrt{a^2 - 1}\right)^n.$$

The recurrence, rather than the radical notation, gives their integer definition. It implies

$$\begin{gathered} Y_n(a) \equiv n \pmod{a-1}, \qquad \nu_2(Y_n(a)) = \nu_2(n) \quad (a \text{ odd},\ n > 0), \\ Y_n(a) \geq (2a-1)^{n-1} \quad (n \geq 1). \end{gathered} \tag{7}$$

The valuation identity follows by repeatedly using $Y_{2m} = 2X_m Y_m$, since every X_m is odd and Y_m is odd exactly for odd m. The last inequality follows from $Y_{n+1} = 2aY_n - Y_{n-1}$ and monotonicity.

By Lemma 2.2, $a = 4t + 1 \geq 9$. Classification of integral Pell solutions gives $F = Y_n(a)$ for a positive n. Since $4t \mid F$, the congruence in (7) yields $4t \mid n$. The exact valuation of $F = dt(2u + 1)$ yields $\nu_2(n) = 19 + \nu_2(t)$. Therefore

$$n \geq dt. \tag{8}$$

The unchanged square-shell constructors are

$$\begin{aligned} I(a,b) &= 2((a+b)^2 + a), & M(a,b) &= I(a,b) + 1, \\ A(a,b) &= 2((a+b)^2 + 3a + 4b + 3), & E(a,b) &= A(a,b) + 1. \end{aligned}$$

For arguments at most m, each output plus one is at most $8(m+1)^2$. All local parameters, and the constants 0 and 1, are strictly below t when $t \geq 2$. The source's authenticated maximum constructor depth is 19. Induction therefore bounds every local label strictly below

$$L = 8^{d-1}t^d, \qquad d = 2^{19}. \tag{9}$$

Combining (7)–(9),

$$F \geq (8t+1)^{n-1} \geq (8t)^{dt-1} \geq L^t. \tag{10}$$

The last ratio is $8^{t-1}/t \geq 1$, including $t = 2$.

Two older numerical uses of $t \geq 16$ deserve explicit replacement. For $\Lambda_t = \operatorname{lcm}(1, \ldots, t)$,

$$\Lambda_t \leq t! \leq t^t, \qquad F \geq L^t \geq t^{2t} \geq \Lambda_t^2.$$

Thus the actual hypothesis of the factorial-free binomial CRT theorem still holds. Also $F \geq L^2 \geq 2^{8d-6} > 2^{32}$, so all coarse lower bounds on the later radix and beta code remain available. Finally the positive endpoint gives $x \geq 2 + (t+1)y$. Since $y \geq F$,

$$2 + t + y + F < x,$$

already for $t \geq 2$, because $(t-1)y > t$. This validates the parent's local bound $2 + t + x + y + F < 2x$.

With these replacements, the finite-product separation theorem uses precisely $F \geq L^t$; the binomial decoder uses $D > 2C$, the canonical common frame, and the retained remainder bound; and the beta and Horn-flow arguments recover a finite $\mathcal{C}_{222}$ refutation. The logical calibration then supplies a finite syntactic ZFC refutation.

2.4 Completeness on a cofinal progression

Starting from a finite calibrated refutation, choose a sufficiently large positive horizon

$$t \equiv 2 \pmod{2^{23}}. \tag{11}$$

This is a cofinal subprogression of (2); it has $\nu_2(t) = 1$, and binary digit 20 of t is zero. All finite proof length and parameter bounds can be met.

Put $n_0 = dt$, $y_0 = Y_{n_0}(4t+1)$, $a_0 = X_{n_0}(4t+1)$, and $b_0 = y_0/(dt)$. The preceding congruence and valuation facts show that b_0 is an odd integer. Since n_0 is even, $a_0 \equiv 1 \pmod 4$.

For every integer $a_0 \equiv 1 \pmod 4$, the map $s \mapsto Y_s(a_0)$ permutes the residue classes modulo each power of two. Here is the needed induction: for $h = 2^r$, the norm and valuation identities give

$$X_h(a_0) \equiv 1 \pmod{2^{2r+2}}, \qquad Y_h(a_0) \equiv 2^r \pmod{2^{r+1}}.$$

The addition formula yields $Y_{s+2^r}(a_0) - Y_s(a_0) \equiv 2^r \pmod{2^{r+1}}$. Thus the two lifts of every index class produce the two different lifts of its image. Starting modulo two proves the assertion, together with periodicity and arbitrarily large indices in a prescribed class.

Choose $\ell > 23$ with $2^\ell > t$. The two odd integers $t/2$ and b_0 are invertible modulo $2^{\ell-20}$. Choose an arbitrarily large positive odd s satisfying

$$(t/2)b_0 Y_s(a_0) \equiv 1 \pmod{2^{\ell-20}}.$$

Then set

$$F = Y_{n_0 s}(4t+1) = y_0 Y_s(a_0), \quad X = X_{n_0 s}(4t+1), \quad Z = tF, \quad u = \frac{F/(dt) - 1}{2}.$$

All witnesses are integral, (3) holds, and $F \equiv 2^{20} \pmod{2^{\ell}}$. The binary supports of F and t are disjoint. The elementary binary-carry criterion consequently gives $B = \binom{F+t}{t}$ odd.

Keep the beta multiplier independent. Choose $2v+1$ to be a positive odd multiple of every odd prime at most t, and sufficiently large that $y = F(2v+1)$ exceeds all finitely many proof-state values. Every prime at most t divides y. If a prime divided both $1+iy$ and $1+jy$, $1 \le i < j \le t+1$, it would divide $j-i$ and be coprime to y, a contradiction. Hence the beta moduli are pairwise coprime. Full factorial divisibility of y is unnecessary.

The witness order is now acyclic: choose beta code x, then B and $N = B^x$, the nonanchor CRT representatives, an admissible even anchor C larger than all other nominal indices, the affine radix D, the true power outputs, and finally the common Pell frame and cap. The retained CRT theorem applies by (10). This constructs a zero from every finite refutation and proves completeness of the changed source.

2.5 From the semantic source to the two literal quartics

The preceding horizon change is an existence-preserving semantic change. It is not asserted to be a bijection with the old square slice. After that change, all remaining transformations are explicit bijections on integer zero sets.

In the parent ordering, replace row 125 by $x_0 - 2 - 4x_{178} = 0$. Delete that row, use the new coordinate $\xi_0 = x_{178}$, and substitute $x_0 = 2 + 4\xi_0$. Keep the other coordinate order, deleting old coordinate 178. For the sparse profile additionally substitute $F = 64\xi_2$ and divide the last growth-divisibility residual by 64. Lemma 2.1 guarantees the inverse $\xi_2 = F/64$ is integral at every zero. All residuals remain quadratic. The final growth residual is

$$\xi_2 - 16384 - 32768\xi_0 - 32768\xi_{208} - 65536\xi_0\xi_{208}. \tag{12}$$

Its square contributes coefficient 2^{32} at $\xi_0^2\xi_{208}^2$. The last coordinate occurs in no other residual, so this coefficient cannot cancel. Exact collection verifies that no coefficient is larger and that precisely 5,486 monomials remain.

For the lower-height companion use $F = 128\xi_2$, whose inverse is likewise integral. A retained residual with coefficient 16384 at ξ_2^2 is used to remove the corresponding square terms from two other rows by integer row additions. With zero-based final row indices, the operations are

$$R_{98} \leftarrow R_{98} + 4R_{135}, \qquad R_{124} \leftarrow R_{124} + 6R_{135},$$

with R_{135} retained. The certificate lists every coefficient. Such a triangular residual-basis operation has an integer inverse; it changes the sum-of-squares polynomial while preserving its zero set. The resulting polynomial has 6,654 monomials and height 2^{30}. Both statements are established by full coefficient collection, not by estimates of input literals.

3 A companion equation of degree 190 in ten positive integers

The compact companion has ten strictly positive integer variables, exact total degree 190, and 119 live binary arithmetic operations (74 multiplications, 24 additions, and 21 subtractions). Its coefficient tuple is unchanged from source 172. This is a positive-domain result, separate from the unrestricted-integer principal construction. It lowers the inspected ten-positive-variable degrees 214, 208, and 206, at a higher arithmetic cost than the 87- and 90-operation alternatives. It does not establish global minimality.

The new step is a transport polynomial $L = UEb^2$ that combines the free-quotient chart with the whole-product center. Transport by bV would leave a degree-197 cross term. Bare b^2 does not have the needed uniform bound when the main Pell index is three. The additional factor UE handles this exceptional index while keeping the cross term below degree 190.

3.1 Complete formula

The independent variables are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta > 0.$$

Use the complete inherited fixed tuple (β, K_0, M_C, M_F), with $\beta = 2^{590069} - 1$. Its four full numerical payloads and finite reconstruction recipe accompany the equation. Define

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha,\\
X &= qw, & Y &= s, & Y' &= qY,\\
k &= \eta + \zeta, & c &= kY + \eta, & A &= Y'(X + 1) + 1,\\
\Delta &= A^2 - 1, & H &= 4A - 5, & v &= X(Y')^2,\\
\ell &= vk, & E &= 1 + \ell(\ell + k), & U &= ic^2,\\
a &= \Delta U^2 E, & F_0 &= 1 + a, & G &= EF_0,\\
f &= 1 + 2a, & Q &= 4\Delta a F_0, & K &= X(3c - 2X) + 2.
\end{aligned}$$

Also set

$$\begin{aligned}
R &= [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J,\\
B &= (K_0 + w)C - F, \qquad b = cT - R, \qquad V = fb - c,\\
e_{\mathrm{clr}} &= 6(k - R - 1 - q^4 Y) - \ell(k^2 - 4),\\
D &= 6q^7, \qquad M = hDH.
\end{aligned}$$

The center is the following division-free polynomial:

$$\begin{aligned}
e_2 &= (kY')^2,\\
S &= (kY')[\ell + k - (kY')(Y' + 1)],\\
\theta &= U^2 b, \qquad R_c = \theta(3E + e_2) - (b - c),\\
S_c &= 4E^2 + 3S^2 + e_2^2,\\
\Omega &= U\{2SR_c + A(4EV - \theta S_c)\}.
\end{aligned}$$

Finally put

$$\begin{aligned}
L &= UEb^2,\\
Z &= hD\{H(n - LB - \Omega) + K\} - H(DB + he_{\mathrm{clr}}),\\
\mathcal{P}_{190} &= Q(Z^2 - 16M^2GV^2) + (16M^2G - Z^2).
\end{aligned}$$

Set $P_{10} = \mathcal{P}_{190}$. The equation is $P_{10} = 0$. All named registers are polynomial abbreviations, not extra quantified variables. There is no variable exponent, quotient, factorial, or hidden predicate in this definition.

3.2 Uniform transport and soundness

The unchanged strong-root and all-index folding argument gives, at a zero, positive integers g, y with

$$G = g^2, \quad QV^2 - (Q-1)y^2 = 1, \quad Z/M = \pm 4gy, \quad c = \psi_A(p), \quad p \geq 3 \text{ odd}.$$

These conclusions precede every sign assertion concerning B, C, R, b, V. Indeed $Q = 4\Delta^2 U^2 G$ and $(Q-1)Z^2 = 16M^2 G(QV^2 - 1)$ force the square class of G by prime valuations; the fixed square factor 16 changes no valuation parity. The retained adjacent-coefficient descent proves the initially rational $y = |Z/M|/(4g)$ integral. The signed folding then gives $c \geq c_3 = 4\Delta + 3$.

The case $b = 0$ would give $0 < y^2 - c^2 = (c^2 - 1)/(Q-1) < 1$, which is impossible. Thus $b \neq 0$, V has its sign, and the auxiliary odd-index classification gives $|V| \geq 4Q - 3$. Since $2c < f$ and $c_3 \leq c$,

$$|b| \geq \frac{|V| - c}{f} \geq 4\Delta f - \frac{c + c_3}{f} > 4\Delta f - 1 > 3\Delta f.$$

This explicitly permits auxiliary index three. Also

$$y < \tfrac{3}{2}|V|, \qquad |V| < \tfrac{3}{2} f|b|, \qquad (AUE)^2 - G = E(U^2 E - 1) > 0,$$

so $g < AUE$.

For the signed center estimate, temporarily regard A as an indeterminate. Then

$$\begin{aligned} E &= e_2 A^2 + e_1 A + e_0, \\ e_1 &= k^2 Y'(1 - 2Y' - 2(Y')^2), \\ e_0 &= 1 + k^2 Y'(Y'+1)[Y'(Y'+1) - 1]. \end{aligned}$$

Let Q_j be the formal polynomial quotient of EV by A^j. Exact polynomial algebra gives

$$\Omega = U(4AEV - 2Q_1 - Q_3/2).$$

Every coefficient of Q_3 is even, because $EV = (b-c)E + 2U^2 b(A^2 - 1)E^2$ and E has A-degree two. The formula above is proof notation for the displayed division-free center.

Put $\delta = (Y'+1)/A < 1/16$. Absolute-coefficient evaluation gives $E^{\#}(A) < 2E$ and

$$(EV)^{\#}(A) < 2E(|b| + c) + 8(A^2 + 1)U^2 E^2 |b| < 10|EV|.$$

For the last inequality use $c + 1 < 2U^2 E$ and $|V| > (f - c)|b| > \Delta U^2 E|b|$; the resulting ratio is at most $8 + 20/\Delta < 10$. Hence

$$|\Omega - 4AUEV| < 10U|EV| \left(\frac{2}{A} + \frac{1}{2A^3}\right) < AUE|V|.$$

In particular Ω has the sign of V and $|\Omega| < 5AUE|V|$. Combining these estimates,

$$4gy + |\Omega| < \frac{33}{2} AUEf|b| < \frac{11A}{2\Delta} L < \frac{L}{4}.$$

The final inequality follows from $A > q^2 \geq 1024$, and is independent of the signs still to be recovered.

For proof only define

$$\rho = \frac{k - R - 1 - vk(k^2 - 4)/6}{q^7}, \quad \sigma = \frac{Y}{q^3}, \quad \kappa = \frac{K}{H}, \quad t = \frac{B}{h}, \quad W = 1 + hL.$$

The literal carrier satisfies

$$Z/M = n + \kappa + \sigma - Wt - \rho - \Omega.$$

Since $k \geq 2$ and $k - R - 1 < b$, one has $\rho < |b|/q^7$ for every integer R. If $B \leq -1$, then $Wt < -L$, and therefore

$$n = \pm 4gy + \Omega - \kappa - \sigma + Wt + \rho < L/4 - L + |b|/q^7 < 0,$$

contrary to $n > 0$. This proves $B \geq 0$, hence $C > 0$.

The authenticated mask now gives $3q < R < q^4 - 2$. The unchanged normalized index and signed auxiliary-index lemmas identify $p = R$. Because Ω is an integer polynomial, $Z/M + \Omega$ is integral; because $W = 1 + hL$, its residue modulo h is unchanged. The source-172 inverse Lucas argument thus gives $m = (R + 1)/2$, followed by the strict Pell projection $X = 2^R$. Only then is q known to be a power of two, licensing the full q^7 divisibility of the inverse correction. The carrier finally yields $q^3 \mid Y$ and $h \mid B$. The retained native reconstruction and the whole coherent tuple's refutation calibration finish soundness. In particular no factor of three is canceled before the dyadic conclusion.

3.3 Completeness and exact degree

A finite refutation supplies the source-172 canonical positive outer data, with $b, V > 0$, integral $t \geq 0$, integral ρ, κ, σ, and $g > \kappa + \sigma + |\rho|$. The center is positive on these data. Choose

$$n = 4gy + \Omega + (1 + hL)t + \rho - \kappa - \sigma > 0.$$

Then $Z/M = 4gy$, so $\mathcal{P}_{190} = 0$. This constructs fresh positive witnesses; it does not assume that an arbitrary integer shear preserves positivity. Both directions are ordinary arguments formalizable in ZFC, using the explicitly retained arithmetic and refutation calibration.

Let Z_0 be the carrier with Ω omitted and $D_{\mathrm{norm}} = \Omega^2 - 16GV^2$. The exact decomposition is

$$\mathcal{P}_{190} = Q[Z_0^2 - 2MZ_0\Omega + M^2 D_{\mathrm{norm}}] + 16M^2G - (Z_0 - M\Omega)^2.$$

In $\mathbb{Z}[A, Y', k, U, b, c]$ with weights $(4, 2, 1, 5, 4, 2)$, the center has 84 monomials and weight 59; the norm defect has 594 monomials and weight 94, with complete highest part

$$[D_{\mathrm{norm}}]_{94} = 8A^6(Y')^8k^8U^6b^2[A^2 - 6(Y')^4].$$

The actual degrees of $(Q, M, G, V, Z_0, \Omega, L)$ are $(72, 12, 46, 36, 41, 59, 27)$. The five summands in the decomposition have bounds $154, 184, 190, 70, 142$, respectively. Put $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The complete actual highest form is

$$[\mathcal{P}_{190}]_{190} = 9 \cdot 2^{11} q_0^{74} w^{18} s^{52} i^{10} k_0^{32} (q_0 - F)^2 (w^2 - 6s^2).$$

Its coefficient at $J^{74}w^{20}s^{52}i^{10}\eta^{32}F^2$ is $9 \cdot 2^{11}\beta^{74} \neq 0$. Thus the degree is exactly 190 for the actual fixed tuple. The independent checker collects all center and defect coefficients, verifies the graph transformation and the leading form, reconstructs all four fixed numerals, and proves genuine input occurrence by nonzero modular derivatives. These finite checks supplement the unbounded proof and do not decide consistency.

For an explicit unrestricted-integer conversion of this companion, replace each positive variable x_j by $1 + a_j^2 + b_j^2 + c_j(c_j + 1)$. The three-square theorem makes this a surjection from $\mathbb{Z}^3$ onto the positive integers: represent $4m + 1$ by three squares and separate its unique odd coordinate. The resulting polynomial has 30 unrestricted variables and degree 380. This corollary is not an unrestricted-arity record.

4 Four oriented power blocks and three integer profiles

The signed growth interface of Section 2 composes with the oriented-pooling theorem of [3]. The standalone growth certificate leaves fifteen ordinary power requests at four bases: $B, D, q, q+1$, with respective multiplicities $1, 1, 3, 10$. Pooling their private certificates and performing checked unit-pivot substitutions gives the following complete polynomial circuits over unrestricted integers.

Variables	Exact degree	Residual squares	Binary operations	Height bound
130	12	70	2475	$< 2^{54}$
116	16	56	2433	$< 2^{84}$
107	32	47	2419	$< 2^{107}$

Binary operations count additions, subtractions, and multiplications in the specified circuit. The last column bounds the absolute coefficients after full expansion; these are rigorous upper bounds, not measured exact heights. The largest literal integer constant is 2^{20}. The first two profiles use the staged partition described below, with different elimination lists; the third uses one four-base pool. Each saves one variable at its matched degree relative to the inspected $131, 117, 108$-variable predecessors. The degree-16 and degree-32 predecessors use smaller literal constants, so the new forms do not improve every size measure.

4.1 The exact identity and the strict rational-error argument

Use A for the common Pell center, distinct from the standalone growth center $4t+1$. The retained frame and anchor supply

$$W \geq 258, \qquad |A| > W^W, \qquad n^2, m_i^2, k_i^2 < W,$$

and the canonical signed Pell pair (X_i, Y_i) at each requested index k_i. Here n is the actual base and m_i its proposed power output. Set $U_i = X_i - AY_i$, using the retained multiplication wire, and, within a block of common base n, define

$$T_n = 2An - n^2 - 1, \qquad H_n = \sum_i Y_i(m_i - X_i), \tag{13}$$

$$E_n = \sum_i ((m_i - U_i)^2 - Y_i^2) - 2nH_n, \qquad \delta_i = m_i - U_i - nY_i. \tag{14}$$

Expansion gives the exact polynomial identity

$$E_n - T_n \sum_i Y_i^2 = \sum_i \delta_i^2. \tag{15}$$

For any selected blocks J, the weighted pool is the single residual

$$R_J = \sum_{j \in J} n_j E_{n_j} \prod_{\ell \in J \setminus \{j\}} T_{n_\ell} - z_J \prod_{j \in J} T_{n_j}. \tag{16}$$

All products are finite. This residual and its circuit contain no divisions or variable exponentiation. Here is a proof of its unbounded arithmetic meaning. Put $b = \lfloor \sqrt{W-1} \rfloor \geq 16$. If $n \neq 0$, then

$$|T_n| \geq 2|A| - W > W^W, \qquad \operatorname{sign}(n/T_n) = \operatorname{sign}(A).$$

For $n = 0$, $T_n = -1$, and the block's weighted contribution is zero. Dividing $R_J = 0$ by its nonzero denominator product and using (15) makes $Q = \sum_{j,i} n_j \delta_i^2 / T_{n_j}$ an integer. Let J_* contain the nonzero bases and put

$$G = \prod_{j \in J_*} n_j^b, \qquad e_i = Gm_i - (G/n_j^b) n_j^{b+k_i} \quad (j \in J_*).$$

These are integers; $G = 1$ for an empty product. Since $T_n \equiv -1 \pmod{n}$, the canonical Pell congruence also permits negative indices, interpreted by modular inverses, and yields $G\delta_i \equiv e_i \pmod{T_{n_j}}$. Therefore

$$Q' = \sum_{j \in J_*} \sum_i n_j e_i^2 / T_{n_j}$$

differs from $G^2 Q$ by an integer. No coprimality between different T_n's is used. There are at most four blocks and fifteen requests, so $|e_i| \le 2b^{5b}$ and

$$\sum_{j,i} |n_j| e_i^2 \le 60 b^{10b+1} < W^{5b+2} \le W^W.$$

The last step uses $5b + 2 \le b^2 + 1 \le W$. Thus $|Q'| < 1$. Its integrality forces $Q' = 0$, and the common sign of every n_j / T_{n_j} forces every $e_i = 0$. Hence every nonzero-base request is exact in $\mathbb{Q}$: $m_i = n_j^{k_i}$. If $|n_j| \ge 2$, integrality further forces $k_i \ge 0$. At the unit bases ± 1, negative indices remain allowed.

Conversely, exact nonzero-base powers give $T_n \mid \delta_i$, then $T_n \mid E_n$, by (15). The integer $z_J = \sum_{j \in J_*} n_j E_{n_j} / T_{n_j}$ completes the pool. Zero-base blocks remain deliberately unconstrained until the bootstrap excludes them.

4.2 The two partitions and the order of sign recovery

The 107-variable profile uses (16) for all four bases. The staged profiles instead use it for B, D, followed by the unweighted residual

$$E_q T_{q+1} + E_{q+1} T_q - z_* T_q T_{q+1}. \tag{17}$$

The unweighted argument is the same integer-error proof with weights one, provided the two denominators have the same sign. Its numerator bound is no larger than the bound above.

This condition is recovered in the required order. The literal horizon and growth-divisibility row make F even and nonzero; C is even. Thus $B\eta = C + F + 1$ makes $B \ne 0$, while $D = 2(C + F + 1) \equiv 2 \pmod 4$ gives $|D| \ge 2$. The weighted block therefore recovers $q = D^x$ and $x \ge 0$. Consequently $q \notin \{0, -1\}$: for $x = 0$ it is one, and otherwise its absolute value is at least two. Hence $q, q + 1$ are nonzero and have the same sign, as do T_q, T_{q+1}. Equation (17) is now legitimate; in the all-base profile the two previously conditional blocks are now active. Every retained power is exact. The endpoint argument in Section 2 then gives $x \ge 1$, uses the literal request q^t to force $t \ge 2$, and recovers all remaining signs. No power at a zero base, nor a nonnegative-index conclusion at a unit base, is used along this route.

4.3 Integral restoration and what the finite checks establish

Pooling removes eight private coordinates and eight rows from the pinned 216-variable source, then adds two pool quotients for the staged form or one for the all-base form. The deleted singleton certificates restore their modulus as T_n and their old quotient as $-\delta_i / T_n$. A deleted common-base pair restores as

$$L_n = E_n / T_n, \qquad K_n = 2H_n + (2A - n) L_n.$$

The exact-power congruence proves that these quotients are integral on zeros. These restoration maps are not asserted to be polynomials on all integer tuples. Conversely, every original certificate supplies

the new integer pool quotients, so the old and pooled systems have the same projection onto retained coordinates.

The final profiles apply respectively 80, 94, and 102 acyclic unit pivots. Each pivot has the form $\varepsilon v + f = 0$, $\varepsilon \in \{1, -1\}$, and is restored by the integer polynomial $v = -\varepsilon f$ in the certified reverse order. This step gives actual polynomial zero-set bijections. The output circuit is the sum of the squares of all remaining residuals.

The independent checker binds the actual 34 frame atoms, 13 principal Pell pairs, 15 requests, historical affine offsets, private-coordinate deletions, and every local polynomial identity and pivot. The complete request table and restoration data are in `pooling/PROOF.md` and `pooling/certificates/`. Nonzero homogeneous specializations prove the exact degrees; coefficient-norm propagation proves the stated height bounds. Both ordinary and optimized-Python verification pass, including fifteen deliberately corrupted profiles that are rejected. These finite checks authenticate the literal equations. The unbounded pooling proof, the signed-horizon soundness and cofinal completeness, and the named inherited logical calibration supply the equivalence with Con(ZFC); no new end-to-end proof-assistant build is claimed.

5 What the reductions establish, and where they stop

5.1 A rejected extra elimination: trace capacity

An attractive attempt would identify the growth-divisibility witness with the beta-spacing witness. In a proposed shifted-horizon variant, write $h = t + 1$, $d = 2^{19}$, and impose

$$F = dh(2v + 1), \qquad y = F(2v + 1).$$

This would eliminate a variable, but would also force

$$y = \frac{F^2}{dh}.$$

For the retained product encoding of pending labels, a state containing two positive labels j_1, j_2 already has weight

$$(1 + j_1 F)(1 + j_2 F) \geq (1 + F)^2.$$

Every beta modulus used at an index $0 \leq i \leq t$, however, would be bounded by

$$1 + (i + 1)y \leq 1 + hy = 1 + \frac{F^2}{d} < (1 + F)^2.$$

The required canonical beta residue cannot hold this state. Making the Pell solution larger does not repair the mismatch: both sides have quadratic dependence on F, and the coefficient on the capacity side is too small. This invalidates the inherited completeness construction for the fused proposal. It is not, by itself, a proof that no different trace presentation could ever satisfy the proposed equation. No 208-variable result is claimed here.

This obstruction separates two jobs that initially looked redundant. The growth coordinate separates products of labels; the beta multiplier must additionally accommodate the full encoded proof states. The latter must remain independently enlargable in the construction we actually use.

5.2 The degree-190 obstruction for this particular center

The compact degree proof identifies a nonzero leading norm-defect term. Consequently optimizing only the transport polynomial, while retaining the current center and all its other definitions, cannot lower

the degree below 190 once the transport cross term is already below 190. This is a statement about that fixed algebraic family, not a lower bound for ten-variable Diophantine equations in general.

A possible next step is a higher-order integral center. It must be analyzed as a whole: canceling one more leading defect term may enlarge the center, and then the cross term with the independent carrier can become dominant. Another direction is to change the positive quotient chart so that the troublesome index-three case is excluded without adding a variable or a comparably costly polynomial factor. Neither direction has been completed in this package.

5.3 Interpreting "simplest"

The principal quartic is attractive when low degree, explicit modest coefficients, and a transparent direct logical source matter. Its coefficient height is exactly 2^{32}, and all 5,486 collected monomials are supplied. The degree-16 and degree-32 profiles trade larger degree for fewer unrestricted integer variables. Their coefficient-height figures are proved upper bounds, not measured exact heights.

The compact equation uses only ten positive variables, but its fixed coefficients are very large: the largest member of its unchanged tuple has 592,884 bits. The 119-operation count charges arithmetic gates, not the cost of arithmetic on integers of that size. Its final collected coefficient height and full monomial count have not been computed. Treating these coefficients as free unit-cost literals would obscure an important part of its description size.

No claim is made that these ideas have never appeared anywhere in the literature. The precise novelty claim supported here is an explicit improvement over the inspected, pinned repository constructions and their seven incoming archives. We supply the transformations, exact equations, and proof obligations needed to assess that claim.

Finally, a proof in ZFC of the displayed biconditionals is compatible with the usual incompleteness phenomenon. The work proves that a zero can be converted to a refutation and conversely; it does not prove that either one exists, or that neither exists. No consistency assumption is used to choose the finite coefficients.

6 Logical calibration and the proof boundary

The direct quartic and pooled constructions encode the numerical calculus $\mathcal{C}_{222}$. Their new arithmetic proofs establish, in both directions, equivalence between an integer zero and a finite refutation in that calculus. The final logical step uses the repository's explicitly named inherited endpoints

> `KEndpoint.c222_iff_zf_refutation`,
> `ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first identifies numerical-calculus refutability with the fixed ZF refutation presentation; the second transfers refutability between the identified ZF and ZFC presentations. Together they supply the stated syntactic ZFC endpoint. The arithmetic reductions in this article preserve that endpoint rather than introducing a new interpretation of Con(ZFC).

The ten-positive-variable construction has a different inherited boundary. It retains the entire authenticated BFS943/radix67 coefficient tuple and the ordinary native/compiler/refutation calibration used by source 172. The selected ordinary proof closure is in `compact/source/upstream172/`. Its `inherited/strict/upstream_lucas/parent/proof_sources/` directory contains the Pell classification and signed-index arguments in `universal_pell.tex`, the uniform projector in `uniform_ratio.md`, the named source theorem in `scaled_native.md`, and the coherent-tuple calibration in `indexed_native.md`.

The local source ledger `compact/source/SELECTED_PROOF_SOURCES.json` and `compact/SOURCE_LINKS.json` identify those retained texts and their immutable repository sources. Reconstructing the coefficient bytes authenticates which instance is used; it does not independently prove its machine or native calibration.

Three kinds of evidence must therefore be distinguished. The new unbounded proofs establish sign recovery, the cofinal existence of witnesses, growth, exact-power interpretation, and the arithmetic restorations. Finite exact checks authenticate the literal equations, substitutions, fixed numerals, degree certificates, and measured size claims. The named inherited calibrations supply the logical meaning of the numerical or native endpoint. Neither finite sampling nor successful coefficient reconstruction replaces an unbounded proof.

All new deductions are ordinary mathematics formalizable in ZFC. This package does not supply a newly compiled proof object for the complete chain, rebuild the entire historical Lean/Rocq import closure, or emit an object-language ZFC derivation. Historical replay statements inside preserved source documents belong to their original releases. The new internal reviews, including `reviews/QUARTIC_ALGEBRA_REVIEW.md` and `reviews/SIGNED_HORIZON_SECOND_REVIEW.md`, are AI-assisted mathematical checks, not external human peer review. No consistency assumption is used to select a coefficient or witness family.

7 Artifacts and reproducibility

The JSON specifications are complete finite definitions of the polynomials. Named gates are shared expressions, not additional quantified variables. The principal artifact locations are as follows.

- The two quartic directories are `quartic/equations/affine209_F64/` and `quartic/equations/affine209_F128_basis/`. Each contains `equation.json`, `expanded_polynomial.json`, `residuals.txt`, `metrics.json`, and `certificate.json`. The exact affine-source and scaling argument is in `quartic/ALGEBRA_PROOF.md`.
- The principal pooled circuits are `pooling/equations/integer116_degree16_staged.json` and `pooling/equations/integer107_degree32_all4.json`. The degree-12 companion is `pooling/equations/integer130_degree12_staged.json`. Their literal source bindings and acyclic restoration certificates are under `pooling/sources/` and `pooling/certificates/`; `pooling/PROOF.md` gives the unbounded arithmetic argument.
- The compact circuit is `compact/equation.json`, with its proof in `compact/PROOF.md`. All center and norm-defect coefficients are recorded in `compact/exact_center_polynomials.json`. The four complete fixed numerical payloads are under `compact/source/coefficients/`; `compact/source/rebuild_coefficients.py` reconstructs them by the printed finite recipe. They are not represented only by checksums.

Python 3.10 or later and its standard library suffice for the exact checks. From the extracted package root, run

```
python3 -B verify_all.py
```

The component entry points can also be run separately:

```
python3 -B quartic/verify.py
python3 -B pooling/verify.py
python3 -B compact/verify.py
```

The same component commands accept Python's `-O` switch before the script name; their acceptance conditions do not depend on removable `assert` statements. The root verifier and retained receipts state which checks were actually replayed.

The quartic verifier independently expands each coordinate substitution and residual-basis identity, collects the sums of squares, and checks every coefficient, support count, and height-attaining quartic monomial. The pooling verifier checks the actual request/frame bindings, cleared denominator identities, and acyclic polynomial restoration maps; exact nonzero homogeneous specializations certify degree. The compact verifier imports no circuit producer: it checks all center gates, the complete norm defect, the actual highest form, and the four fixed numerals by independent finite reconstruction. Nonzero modular derivatives certify genuine occurrence of the circuit inputs. These modular checks are finite algebraic certificates, not evidence for or against an integer zero.

The main article source is `report/main.tex`; every included `.tex` fragment is supplied. With a standard TeX Live installation, change into the `report/` directory and run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

PDF metadata can vary between TeX installations. The literal equation data, integer coefficients, and exact certificates are the stable mathematical definitions.

For repository integration, retain this package as an additive submission and cite the specific profile, domain, degree, and cost being compared. Preserve the source notices and the distinction between the two calibration routes. The results support the measured comparisons in this article; they do not establish a global simplicity minimum or historical priority.

References

[1] Vladimir Reshetnikov and contributors, *ProveIt*, snapshot `d123610bc98fef2006bff1c9466bc3f1221003e7`. https://github.com/VladimirReshetnikov/ProveIt. The exact repository paths, archive SHA-256 hashes, and copied-file hashes are recorded in the accompanying provenance data.

[2] ProveIt incoming research package, *A divisibility-forced Pell growth certificate for Diophantine consistency*, `ConZFC_Pell_Growth_210_2026-10-25.zip`, 2026. The new arithmetic uses its `proofs/growth_proof.md`, affine-radix proof, literal 210-variable equation, and named source dependencies.

[3] ProveIt incoming research package, *Signed horizons and residue-encoded Pell witnesses*, `ConZFC_Signed_Horizons_and_Residue_Carriers_2026-10-25.zip`, 2026. Its oriented-pooling theorem and source-bound projection method are retained.

[4] ProveIt package 172, *Linear trace, free quotient, and inverse-Lucas composition*, `172-pooled-powers-round21/`, part 2, `scalar/positive10/`, 2026. The exact coherent BFS943/radix67 numerical tuple and the complete degree-214 parent are retained in the archive.

[5] ProveIt package 178, *Whole-product polynomial centers*, `178-polynomial-centers/`, 2026. The formal-quotient center identity is independently collected again here.

[6] ProveIt incoming research package, *Affine horizons and centered certificates for Diophantine consistency*, `ConZFC_Affine_Horizons_Centered_Certificates_2026-10-26.zip`, 2026. Its ten-positive-variable degree-206 result is included in the comparison.

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A The complete 209-variable quartic

The unknowns are exactly $\xi_0, \dots, \xi_{208} \in \mathbb{Z}$. The following 149 residuals define $Q_{209} = \sum_{j=0}^{148} R_j^2$. Products are ordinary multiplication; every residual has degree at most two. The independent archive checker parses every displayed coefficient and monomial below and compares them with the literal JSON equation. The fully collected 5,486-term expansion is supplied as data.

The lower-height companion is also completely specified by this list. First replace every ξ_2 by $2\xi_2$, then divide residual 148 by two (its coefficients after substitution are all even). Call the resulting residuals $\widetilde{R}_j$. Set

$$S_{98} = \widetilde{R}_{98} + 4\widetilde{R}_{135}, \qquad S_{124} = \widetilde{R}_{124} + 6\widetilde{R}_{135},$$

and $S_j = \widetilde{R}_j$ for every other index. The second equation is $\sum_{j=0}^{148} S_j^2 = 0$. This transformation gives its full 6,654-term expansion without any additional choice of constants.

$$\begin{aligned}
R_0 &= 2 + 4\xi_0 - 4\xi_0\xi_4 - 3\xi_4 + \xi_7 \\
R_1 &= -64\xi_2 - \xi_4\xi_5 + \xi_6\xi_7 \\
R_2 &= -1 + \xi_1 - \xi_6\xi_7 \\
R_3 &= -128\xi_2\xi_{205} - \xi_3\xi_{18} + \xi_4 + 2\xi_{16} \\
R_4 &= -4 - 3\xi_8 - \xi_8^2 - 2\xi_8\xi_9 - 4\xi_9 - \xi_9^2 + \xi_{19} \\
R_5 &= \xi_8 - \xi_8^2 - 8\xi_8\xi_{19} - 4\xi_8\xi_{20} - 16\xi_{19}^2 - 16\xi_{19}\xi_{20} \\
&\quad - 4\xi_{20}^2 + \xi_{21} \\
R_6 &= 1 + 2\xi_{21} - 4\xi_{21}^2 + \xi_{22} \\
R_7 &= -2 - \xi_9^2 - 2\xi_9\xi_{10} - \xi_{10}^2 + \xi_{23} \\
R_8 &= 1 - \xi_8^2 - 2\xi_8\xi_{10} - \xi_{10}^2 + \xi_{24} \\
R_9 &= -29 - 61\xi_8 - 35\xi_8^2 - 96\xi_8\xi_{10} - 24\xi_8\xi_{19} - 4\xi_8\xi_{23} - 24\xi_8\xi_{24} \\
&\quad + 16\xi_9 - 2\xi_9^2 - 8\xi_9\xi_{23} - 96\xi_{10} - 64\xi_{10}^2 - 32\xi_{10}\xi_{19} \\
&\quad - 32\xi_{10}\xi_{24} - 28\xi_{19} - 4\xi_{19}^2 - 8\xi_{19}\xi_{24} + 8\xi_{23} - 4\xi_{23}^2 - 24\xi_{24} \\
&\quad - 4\xi_{24}^2 + \xi_{25} \\
R_{10} &= -4 + 10\xi_8 - 2\xi_8^2 - 8\xi_8\xi_9 - 8\xi_8\xi_{23} - 4\xi_{25}^2 + \xi_{26} - 2\xi_{43} \\
R_{11} &= -4 - 16\xi_{10}^2 - 16\xi_{10}\xi_{23} - 16\xi_{10}\xi_{24} - 4\xi_{23}^2 - 8\xi_{23}\xi_{24} - 4\xi_{24}^2 \\
&\quad + \xi_{27} \\
R_{12} &= -7 + 24\xi_{10} - 16\xi_{10}^2 - 16\xi_{10}\xi_{19} - 32\xi_{10}\xi_{24} - 16\xi_{10}\xi_{27} + 10\xi_{19} \\
&\quad - 4\xi_{19}^2 - 16\xi_{19}\xi_{24} - 8\xi_{19}\xi_{27} + 24\xi_{24} - 16\xi_{24}^2 - 16\xi_{24}\xi_{27} \\
&\quad + 12\xi_{27} - 4\xi_{27}^2 + \xi_{28}
\end{aligned}$$

$$
\begin{aligned}
R_{13} = {} & -114-127\xi_{8}-37\xi_{8}^{2}-24\xi_{8}\xi_{9}-112\xi_{8}\xi_{10}-4\xi_{8}\xi_{23}\\
& -24\xi_{8}\xi_{24}-67\xi_{9}-37\xi_{9}^{2}-112\xi_{9}\xi_{10}-24\xi_{9}\xi_{23}-4\xi_{9}\xi_{24}\\
& -224\xi_{10}-128\xi_{10}^{2}-32\xi_{10}\xi_{23}-32\xi_{10}\xi_{24}-16\xi_{23}-4\xi_{23}^{2}-40\xi_{24}\\
& -4\xi_{24}^{2}+\xi_{29}\\
R_{14} = {} & -30-14\xi_{8}-2\xi_{8}^{2}-24\xi_{8}\xi_{9}-32\xi_{8}\xi_{10}-8\xi_{8}\xi_{23}-96\xi_{9}\\
& -72\xi_{9}^{2}-192\xi_{9}\xi_{10}-48\xi_{9}\xi_{23}-128\xi_{10}-128\xi_{10}^{2}-64\xi_{10}\xi_{23}\\
& -32\xi_{23}-8\xi_{23}^{2}-4\xi_{29}^{2}+\xi_{30}\\
R_{15} = {} & -74-42\xi_{8}+8\xi_{8}\xi_{10}+16\xi_{8}\xi_{23}-84\xi_{9}+16\xi_{9}\xi_{10}+32\xi_{9}\xi_{23}\\
& +29\xi_{10}-3\xi_{10}^{2}-4\xi_{10}\xi_{19}-8\xi_{10}\xi_{23}+21\xi_{19}-8\xi_{19}\xi_{23}+40\xi_{23}\\
& -4\xi_{23}^{2}+\xi_{31}-\xi_{59}\\
R_{16} = {} & -131-184\xi_{8}-64\xi_{8}^{2}-160\xi_{8}\xi_{10}-64\xi_{8}\xi_{23}-32\xi_{8}\xi_{24}\\
& -32\xi_{8}\xi_{31}-226\xi_{10}-100\xi_{10}^{2}-80\xi_{10}\xi_{23}-40\xi_{10}\xi_{24}-40\xi_{10}\xi_{31}\\
& -88\xi_{23}-16\xi_{23}^{2}-16\xi_{23}\xi_{24}-16\xi_{23}\xi_{31}-46\xi_{24}-4\xi_{24}^{2}\\
& -8\xi_{24}\xi_{31}-44\xi_{31}-4\xi_{31}^{2}+\xi_{32}\\
R_{17} = {} & -4\xi_{19}^{2}+\xi_{20}\\
R_{18} = {} & \xi_{8}-\xi_{8}^{2}+8\xi_{8}\xi_{19}-4\xi_{8}\xi_{20}-16\xi_{19}^{2}+16\xi_{19}\xi_{20}\\
& -4\xi_{20}^{2}+\xi_{33}\\
R_{19} = {} & 1+2\xi_{33}-4\xi_{33}^{2}+\xi_{34}\\
R_{20} = {} & -\xi_{9}^{2}-4\xi_{9}\xi_{34}-4\xi_{34}^{2}+\xi_{35}\\
R_{21} = {} & -44-52\xi_{8}-16\xi_{8}^{2}-64\xi_{8}\xi_{9}+32\xi_{8}\xi_{19}-104\xi_{9}-64\xi_{9}^{2}\\
& +64\xi_{9}\xi_{19}+50\xi_{19}-16\xi_{19}^{2}+\xi_{36}\\
R_{22} = {} & 8\xi_{9}-62\xi_{9}^{2}-92\xi_{9}\xi_{10}-32\xi_{9}\xi_{23}-32\xi_{9}\xi_{36}+6\xi_{10}\\
& -34\xi_{10}^{2}-24\xi_{10}\xi_{23}-24\xi_{10}\xi_{36}-4\xi_{23}^{2}-8\xi_{23}\xi_{36}+4\xi_{36}\\
& -4\xi_{36}^{2}+\xi_{37}\\
R_{23} = {} & -90-152\xi_{8}-64\xi_{8}^{2}-96\xi_{8}\xi_{10}-32\xi_{8}\xi_{24}-32\xi_{8}\xi_{37}-114\xi_{10}\\
& -36\xi_{10}^{2}-24\xi_{10}\xi_{24}-24\xi_{10}\xi_{37}-38\xi_{24}-4\xi_{24}^{2}-8\xi_{24}\xi_{37}\\
& -36\xi_{37}-4\xi_{37}^{2}+\xi_{38}\\
R_{24} = {} & -\xi_{10}^{2}-2\xi_{10}\xi_{11}-\xi_{11}^{2}+\xi_{39}\\
R_{25} = {} & -7+8\xi_{8}-3\xi_{8}^{2}-8\xi_{8}\xi_{9}-8\xi_{8}\xi_{23}-8\xi_{8}\xi_{24}-4\xi_{8}\xi_{39}\\
& -8\xi_{9}\xi_{24}-2\xi_{10}^{2}-8\xi_{10}\xi_{39}-8\xi_{23}\xi_{24}+6\xi_{24}-4\xi_{24}^{2}\\
& -4\xi_{39}^{2}+\xi_{40}-\xi_{43}\\
R_{26} = {} & -16\xi_{8}^{2}-36\xi_{8}\xi_{10}-32\xi_{8}\xi_{24}-40\xi_{8}\xi_{39}-16\xi_{8}\xi_{40}-4\xi_{10}\\
& -22\xi_{10}^{2}-32\xi_{10}\xi_{24}-48\xi_{10}\xi_{39}-16\xi_{10}\xi_{40}-2\xi_{24}-16\xi_{24}^{2}\\
& -32\xi_{24}\xi_{39}-16\xi_{24}\xi_{40}-4\xi_{39}-24\xi_{39}^{2}-16\xi_{39}\xi_{40}-4\xi_{40}^{2}\\
& +\xi_{41}\\
R_{27} = {} & \xi_{10}-25\xi_{10}^{2}-40\xi_{10}\xi_{24}-20\xi_{10}\xi_{27}-16\xi_{24}^{2}-16\xi_{24}\xi_{27}\\
& -4\xi_{27}^{2}+\xi_{42}\\
R_{28} = {} & -1+8\xi_{9}-4\xi_{9}^{2}-8\xi_{9}\xi_{23}+8\xi_{23}-4\xi_{23}^{2}+\xi_{43}\\
R_{29} = {} & -1-16\xi_{9}^{2}-8\xi_{9}\xi_{10}-32\xi_{9}\xi_{23}-16\xi_{9}\xi_{43}-\xi_{10}^{2}-8\xi_{10}\xi_{23}\\
& -4\xi_{10}\xi_{43}-16\xi_{23}^{2}-16\xi_{23}\xi_{43}-4\xi_{43}^{2}+\xi_{44}
\end{aligned}
$$

$$R_{30} = 3 + \xi_{9} - 2\xi_{9}\xi_{10} - 4\xi_{9}\xi_{44} - 3\xi_{10}^{2} - 8\xi_{10}\xi_{44} - \xi_{23} - 4\xi_{44}^{2} + \xi_{45}$$

$$R_{31} = -1 - 4\xi_{45} - 4\xi_{45}^{2} + \xi_{46}$$

$$R_{32} = -1 - \xi_{10}^{2} - 8\xi_{10}\xi_{45} - 4\xi_{10}\xi_{46} - 16\xi_{45}^{2} - 16\xi_{45}\xi_{46} - 4\xi_{46}^{2} + \xi_{47}$$

$$R_{33} = 2 + \xi_{9} - \xi_{9}^{2} - 4\xi_{9}\xi_{10} - 4\xi_{9}\xi_{47} - 4\xi_{10}^{2} - 8\xi_{10}\xi_{47} - 4\xi_{47}^{2} + \xi_{48}$$

$$R_{34} = 1 - 4\xi_{42}^{2} - 8\xi_{42}\xi_{48} - 4\xi_{48}^{2} + \xi_{49}$$

$$R_{35} = 1 + \xi_{8} - \xi_{8}^{2} - 8\xi_{8}\xi_{42} - 4\xi_{8}\xi_{49} - 16\xi_{42}^{2} - 16\xi_{42}\xi_{49} - 4\xi_{49}^{2} + \xi_{50}$$

$$R_{36} = -1 - 4\xi_{50} - 4\xi_{50}^{2} + \xi_{51}$$

$$R_{37} = -1 - \xi_{10}^{2} - 8\xi_{10}\xi_{50} - 4\xi_{10}\xi_{51} - 16\xi_{50}^{2} - 16\xi_{50}\xi_{51} - 4\xi_{51}^{2} + \xi_{52}$$

$$R_{38} = 2 + \xi_{8} - \xi_{8}^{2} - 4\xi_{8}\xi_{10} - 4\xi_{8}\xi_{52} - 4\xi_{10}^{2} - 8\xi_{10}\xi_{52} - 4\xi_{52}^{2} + \xi_{53}$$

$$R_{39} = -10 - 12\xi_{53} - 4\xi_{53}^{2} + \xi_{54}$$

$$R_{40} = -19 + 4\xi_{9} - 4\xi_{9}^{2} - 8\xi_{9}\xi_{23} - 16\xi_{9}\xi_{53} - 8\xi_{9}\xi_{54} + 4\xi_{23} - 4\xi_{23}^{2} - 16\xi_{23}\xi_{53} - 8\xi_{23}\xi_{54} - 16\xi_{53} - 24\xi_{53}^{2} - 16\xi_{53}\xi_{54} + 6\xi_{54} - 4\xi_{54}^{2} + \xi_{55}$$

$$R_{41} = -10 + \xi_{9} - 25\xi_{9}^{2} - 40\xi_{9}\xi_{23} - 20\xi_{9}\xi_{55} - 16\xi_{23}^{2} - 16\xi_{23}\xi_{55} - 12\xi_{48} - 4\xi_{48}^{2} - 4\xi_{55}^{2} + \xi_{56}$$

$$R_{42} = 21 + 24\xi_{48} - 8\xi_{48}^{2} - 16\xi_{48}\xi_{56} - 2\xi_{56} - 4\xi_{56}^{2} + \xi_{57}$$

$$R_{43} = -1 + 4\xi_{8} + 8\xi_{9} + \xi_{10} - \xi_{10}^{2} - 4\xi_{10}\xi_{57} - 2\xi_{19} - 4\xi_{57}^{2} + \xi_{58} + 12\xi_{59} - 4\xi_{59}^{2}$$

$$R_{44} = -22 - 38\xi_{8} - 16\xi_{8}^{2} - 64\xi_{8}\xi_{9} + 8\xi_{8}\xi_{10} + 16\xi_{8}\xi_{19} - 76\xi_{9} - 64\xi_{9}^{2} + 16\xi_{9}\xi_{10} + 32\xi_{9}\xi_{19} + 11\xi_{10} - \xi_{10}^{2} - 4\xi_{10}\xi_{19} + 19\xi_{19} - 4\xi_{19}^{2} + \xi_{59}$$

$$R_{45} = -4 + 8\xi_{8} + 16\xi_{9} - 4\xi_{19} - 4\xi_{58}^{2} + 24\xi_{59} - 8\xi_{59}^{2} + \xi_{60}$$

$$R_{46} = -1 - 4\xi_{60} - 4\xi_{60}^{2} + \xi_{61}$$

$$R_{47} = -64\xi_{2}\xi_{13} + \xi_{62}$$

$$R_{48} = \xi_{12} - \xi_{13} - 2\xi_{22}\xi_{62} - \xi_{63}$$

$$R_{49} = \xi_{12} - \xi_{13} - 2\xi_{26}\xi_{62} - \xi_{64}$$

$$R_{50} = \xi_{12} - \xi_{13} - 2\xi_{28}\xi_{62} - 2\xi_{62} - \xi_{65}$$

$$R_{51} = \xi_{12} - \xi_{13} - 2\xi_{30}\xi_{62} - \xi_{66}$$

$$R_{52} = \xi_{12} - \xi_{13} - 2\xi_{32}\xi_{62} + 2\xi_{62} - \xi_{67}$$

$$R_{53} = \xi_{12} - \xi_{13} - 4\xi_{34}\xi_{62} - 2\xi_{35}\xi_{62} - \xi_{68}$$

$$R_{54} = \xi_{12} - \xi_{13} - 2\xi_{38}\xi_{62} - \xi_{69}$$

$$R_{55} = 2\xi_{28}\xi_{62} - 2\xi_{41}\xi_{62} - 2\xi_{62} + \xi_{65} - \xi_{70}$$

$$R_{56} = -\xi_{8}\xi_{9} + \xi_{8}\xi_{10} + \xi_{9}\xi_{10} - \xi_{10}^{2} + \xi_{71}$$

$$R_{57} = 1 + 2\xi_{11} + \xi_{11}^2 - \xi_{71}^2 + \xi_{72}$$

$$\begin{aligned} R_{58} = & -\xi_{12}^2 + 2\xi_{12}\xi_{13} + 2\xi_{12}\xi_{74} - \xi_{13}^2 - 2\xi_{13}\xi_{74} - \xi_{72}^2 \\ & + \xi_{73} - \xi_{74}^2 \end{aligned}$$

$$R_{59} = -4\xi_{60}\xi_{62} - 2\xi_{61}\xi_{62} + \xi_{74}$$

$$R_{60} = -\xi_{9}\xi_{62} - \xi_{13} + \xi_{75}\xi_{76} + \xi_{77}$$

$$R_{61} = -1 + 512\xi_2 + 256\xi_2\xi_8 + 512\xi_2\xi_9 - 128\xi_2\xi_{19} + \xi_{75}$$

$$R_{62} = -\xi_{8}\xi_{200} - \xi_{12} + \xi_{76}$$

$$R_{63} = \xi_{9}\xi_{200} - 2\xi_{19}\xi_{62} + 2\xi_{32}\xi_{62} + \xi_{67} + \xi_{78}$$

$$R_{64} = -\xi_{12}\xi_{63} + \xi_{13}\xi_{63} + \xi_{79}$$

$$R_{65} = -\xi_{64}\xi_{79} + \xi_{80}$$

$$R_{66} = -\xi_{65}\xi_{80} + \xi_{81}$$

$$R_{67} = -\xi_{66}\xi_{81} + \xi_{82}$$

$$R_{68} = -\xi_{67}\xi_{82} + \xi_{83}$$

$$R_{69} = -\xi_{68}\xi_{83} + \xi_{84}$$

$$R_{70} = -\xi_{69}\xi_{84} + \xi_{85}$$

$$R_{71} = -\xi_{70}\xi_{85} + \xi_{86}$$

$$R_{72} = -\xi_{77}\xi_{86} + \xi_{87}$$

$$R_{73} = -\xi_{78}\xi_{87} + \xi_{88}$$

$$R_{74} = -\xi_{88}^2 + \xi_{89}$$

$$R_{75} = -\xi_{4} - 2\xi_{4}\xi_{16} + 2\xi_{16} + \xi_{90}$$

$$R_{76} = -\xi_{14}\xi_{90} + \xi_{91}$$

$$\begin{aligned} R_{77} = & -2\xi_1^2 + 2\xi_1\xi_{12} + 2\xi_1\xi_{13} + 2\xi_1\xi_{91} + 2\xi_1\xi_{92} - \xi_{12}^2 \\ & - 2\xi_{12}\xi_{91} - \xi_{13}^2 - 2\xi_{13}\xi_{92} + \xi_{17}\xi_{93} - \xi_{73}\xi_{89} - \xi_{91}^2 \\ & - \xi_{92}^2 \end{aligned}$$

$$R_{78} = -\xi_{4}\xi_{15} + \xi_{15} - \xi_{15}\xi_{90} + \xi_{92}$$

$$R_{79} = -\xi_{94}\xi_{97} + \xi_{97} + \xi_{99}$$

$$R_{80} = -\xi_{3}\xi_{120} + \xi_{95}\xi_{192} - \xi_{98} - \xi_{99}\xi_{120}$$

$$R_{81} = \xi_{100}\xi_{192} - \xi_{101}\xi_{123} - \xi_{102}$$

$$R_{82} = \xi_{103}\xi_{192} - \xi_{104}\xi_{123} - \xi_{105}$$

$$R_{83} = \xi_{106}\xi_{192} - \xi_{107}\xi_{123} - \xi_{108}$$

$$R_{84} = \xi_{109}\xi_{192} - \xi_{110}\xi_{123} - \xi_{111}$$

$$R_{85} = \xi_{112}\xi_{193} - \xi_{114}\xi_{123} - \xi_{115}$$

$$R_{86} = \xi_{116}\xi_{193} - \xi_{117}\xi_{123} - \xi_{118}$$

$$R_{87} = -\xi_{17}\xi_{120} + \xi_{123}$$

$$R_{88} = \xi_{119} - \xi_{121}\xi_{123} - \xi_{122}$$

$$R_{89} = -\xi_{123}\xi_{125} + \xi_{124} - \xi_{126}$$

$$R_{90} = -\xi_{123}\xi_{128} + \xi_{127}\xi_{192} - \xi_{129}$$

$$R_{91} = \xi_{130} - 2\xi_{130}\xi_{132} + \xi_{133}$$

$$R_{92} = -\xi_{130}\xi_{133} - \xi_{133} + \xi_{134}$$

$$R_{93} = -1 + \xi_{131}^2 + \xi_{133}^2 - \xi_{134}^2$$

$$R_{94} = 1 - \xi_{131}\xi_{136} - \xi_{135} + \xi_{137}$$

$$R_{95} = -1 + 2\xi_{135} - \xi_{135}^2 + 2\xi_{135}\xi_{137} + \xi_{136}^2$$

$$R_{96} = 1 + \xi_3^2 - 2\xi_3\xi_{131} + \xi_{138}$$

$$R_{97} = -\xi_3\xi_{170} + \xi_{17} + \xi_{138}\xi_{139} - \xi_{169}$$

$$R_{98} = -5 - 512\xi_2 - 16384\xi_2^2 - 1024\xi_2\xi_{16} + 256\xi_2\xi_{131} - 16\xi_{16} - 16\xi_{16}^2 + 8\xi_{16}\xi_{131} + 4\xi_{131} - \xi_{140}$$

$$R_{99} = -1 - 128\xi_2\xi_{170} - 4\xi_{16}\xi_{170} + \xi_{94} + \xi_{140}\xi_{141} - \xi_{169} - 2\xi_{170}$$

$$R_{100} = -\xi_{131}\xi_{143} - \xi_{142} + \xi_{144}$$

$$R_{101} = -1 - \xi_{142}^2 + 2\xi_{142}\xi_{144} + \xi_{143}^2$$

$$R_{102} = -\xi_{131}\xi_{146} - \xi_{145} + \xi_{147}$$

$$R_{103} = -1 - \xi_{145}^2 + 2\xi_{145}\xi_{147} + \xi_{146}^2$$

$$R_{104} = -\xi_{131}\xi_{149} - \xi_{148} + \xi_{150}$$

$$R_{105} = -1 - \xi_{148}^2 + 2\xi_{148}\xi_{150} + \xi_{149}^2$$

$$R_{106} = -\xi_{131}\xi_{152} - \xi_{151} + \xi_{153}$$

$$R_{107} = -1 - \xi_{151}^2 + 2\xi_{151}\xi_{153} + \xi_{152}^2$$

$$R_{108} = -\xi_{131}\xi_{155} - \xi_{154} + \xi_{156}$$

$$R_{109} = -1 - \xi_{154}^2 + 2\xi_{154}\xi_{156} + \xi_{155}^2$$

$$R_{110} = -\xi_{131}\xi_{158} - \xi_{157} + \xi_{159}$$

$$R_{111} = -1 - \xi_{157}^2 + 2\xi_{157}\xi_{159} + \xi_{158}^2$$

$$R_{112} = 1 - \xi_{131}\xi_{161} - \xi_{160} + \xi_{162}$$

$$R_{113} = -1 + 2\xi_{160} - \xi_{160}^2 + 2\xi_{160}\xi_{162} + \xi_{161}^2$$

$$R_{114} = -\xi_{131}\xi_{164} - \xi_{163} + \xi_{165}$$

$$R_{115} = -1 - \xi_{163}^2 + 2\xi_{163}\xi_{165} + \xi_{164}^2$$

$$R_{116} = -\xi_{131}\xi_{167} - \xi_{166} + \xi_{168}$$

$$R_{117} = -1 - \xi_{166}^2 + 2\xi_{166}\xi_{168} + \xi_{167}^2$$

$$R_{118} = 1 - \xi_{131}\xi_{170} - \xi_{169} + \xi_{171}$$

$$R_{119} = -1 + 2\xi_{169} - \xi_{169}^2 + 2\xi_{169}\xi_{171} + \xi_{170}^2$$

$$R_{120} = -\xi_{131}\xi_{173} - \xi_{172} + \xi_{174}$$

$$R_{121} = -1 - \xi_{172}^2 + 2\xi_{172}\xi_{174} + \xi_{173}^2$$

$$R_{122} = -\xi_{131}\xi_{176} - \xi_{175} + \xi_{177}$$

$$R_{123} = -1 - \xi_{175}^2 + 2\xi_{175}\xi_{177} + \xi_{176}^2$$

$$
\begin{aligned}
R_{124} = {} & -28-72\xi_0-64\xi_0^2-512\xi_0\xi_2-\xi_1^2-768\xi_2-24576\xi_2^2\\
& -1024\xi_2\xi_{16}-\xi_3^2-\xi_4^2-\xi_8^2-\xi_9^2-\xi_{10}^2\\
& -\xi_{11}^2-\xi_{12}^2-\xi_{13}^2-\xi_{14}^2-\xi_{15}^2-16\xi_{16}\\
& -20\xi_{16}^2-\xi_{17}^2+2\xi_{94}-2\xi_{94}^2-\xi_{95}^2-\xi_{96}^2\\
& -\xi_{100}^2-\xi_{103}^2-\xi_{106}^2-\xi_{109}^2-\xi_{112}^2-\xi_{113}^2\\
& -\xi_{116}^2-\xi_{119}^2-\xi_{120}^2-\xi_{124}^2-\xi_{127}^2+\xi_{130}
\end{aligned}
$$

$$R_{125} = -\xi_{131}\xi_{179}+\xi_{188}$$

$$R_{126} = -1+\xi_{178}^2+\xi_{179}^2-\xi_{188}^2$$

$$R_{127} = -\xi_{181}\xi_{182}+\xi_{189}$$

$$R_{128} = -1+\xi_{180}^2+\xi_{181}^2-\xi_{189}^2$$

$$R_{129} = -\xi_{176}^2+\xi_{190}$$

$$R_{130} = \xi_{179}-2\xi_{184}\xi_{190}-\xi_{190}$$

$$R_{131} = -1-4\xi_{179}-8\xi_{179}\xi_{183}+\xi_{182}$$

$$R_{132} = -\xi_{131}-\xi_{178}\xi_{185}+\xi_{182}$$

$$R_{133} = -\xi_{177}-\xi_{178}\xi_{186}+\xi_{180}$$

$$R_{134} = -2\xi_{16}-4\xi_{176}\xi_{187}+\xi_{181}$$

$$
\begin{aligned}
R_{135} = {} & 17+32\xi_0+32\xi_0^2+512\xi_0\xi_2-8\xi_0\xi_{136}-8\xi_0\xi_{143}+\xi_1^2\\
& -2\xi_1\xi_{170}+256\xi_2+4096\xi_2^2-128\xi_2\xi_{143}+\xi_4^2-2\xi_4\xi_{161}\\
& +\xi_8^2-2\xi_8\xi_{146}+\xi_9^2-2\xi_9\xi_{149}+\xi_{10}^2-2\xi_{10}\xi_{152}\\
& +\xi_{11}^2-2\xi_{11}\xi_{155}+\xi_{12}^2-2\xi_{12}\xi_{158}+\xi_{13}^2-2\xi_{13}\xi_{164}\\
& +\xi_{14}^2-2\xi_{14}\xi_{167}+\xi_{15}^2-2\xi_{15}\xi_{173}-4\xi_{16}\xi_{176}-\xi_{131}\xi_{191}\\
& -4\xi_{136}+\xi_{137}^2-4\xi_{143}-2\xi_{144}+\xi_{144}^2-2\xi_{147}+\xi_{147}^2-2\xi_{150}\\
& +\xi_{150}^2-2\xi_{153}+\xi_{153}^2-2\xi_{156}+\xi_{156}^2-2\xi_{159}+\xi_{159}^2\\
& +\xi_{162}^2-2\xi_{165}+\xi_{165}^2-2\xi_{168}+\xi_{168}^2+\xi_{171}^2-2\xi_{174}\\
& +\xi_{174}^2+2\xi_{176}^2-2\xi_{177}+\xi_{177}^2+\xi_{191}-\xi_{194}^2-\xi_{195}^2\\
& -\xi_{196}-\xi_{196}^2
\end{aligned}
$$

$$R_{136} = -\xi_{96}\xi_{192}+\xi_{120}$$

$$R_{137} = -\xi_{113}\xi_{193}+\xi_{120}$$

$$
\begin{aligned}
R_{138} = {} & -1-4\xi_{16}^2-\xi_{136}^2-\xi_{143}^2-\xi_{146}^2-\xi_{149}^2-\xi_{152}^2\\
& -\xi_{155}^2-\xi_{158}^2-\xi_{161}^2-\xi_{164}^2-\xi_{167}^2-\xi_{170}^2\\
& -\xi_{173}^2+\xi_{176}^2-\xi_{194}^2-\xi_{195}^2-\xi_{196}-\xi_{196}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{139} = {} & -3-4\xi_{98}^2-4\xi_{102}^2-4\xi_{105}^2-4\xi_{108}^2-4\xi_{111}^2-4\xi_{115}^2\\
& -4\xi_{118}^2+\xi_{120}^2-4\xi_{122}^2-4\xi_{126}^2-4\xi_{129}^2-\xi_{197}^2\\
& -\xi_{198}^2+3\xi_{199}-\xi_{199}^2
\end{aligned}
$$

$$R_{140} = -64\xi_2\xi_{12}+\xi_{200}$$

$$
\begin{aligned}
R_{141} = {} & \xi_{94}\xi_{202}-2\xi_{95}\xi_{143}-2\xi_{100}\xi_{146}-2\xi_{103}\xi_{149}-2\xi_{106}\xi_{152}-2\xi_{109}\xi_{155}\\
& -2\xi_{112}\xi_{158}-2\xi_{116}\xi_{164}-2\xi_{119}\xi_{167}-2\xi_{124}\xi_{173}-2\xi_{127}\xi_{176}-2\xi_{131}\xi_{202}\\
& +2\xi_{143}\xi_{144}+2\xi_{146}\xi_{147}+2\xi_{149}\xi_{150}+2\xi_{152}\xi_{153}+2\xi_{155}\xi_{156}+2\xi_{158}\xi_{159}\\
& +2\xi_{164}\xi_{165}+2\xi_{167}\xi_{168}+2\xi_{173}\xi_{174}+2\xi_{176}\xi_{177}+\xi_{201}
\end{aligned}
$$

$$
\begin{aligned}
R_{142} = &-10-\xi_{94}\xi_{201}+\xi_{95}^2-2\xi_{95}\xi_{142}+\xi_{100}^2-2\xi_{100}\xi_{145}+\xi_{103}^2\\
&-2\xi_{103}\xi_{148}+\xi_{106}^2-2\xi_{106}\xi_{151}+\xi_{109}^2-2\xi_{109}\xi_{154}+\xi_{112}^2\\
&-2\xi_{112}\xi_{157}+\xi_{116}^2-2\xi_{116}\xi_{163}+\xi_{119}^2-2\xi_{119}\xi_{166}+\xi_{124}^2\\
&-2\xi_{124}\xi_{172}+\xi_{127}^2-2\xi_{127}\xi_{175}+2\xi_{142}\xi_{144}+2\xi_{145}\xi_{147}+2\xi_{148}\xi_{150}\\
&+2\xi_{151}\xi_{153}+2\xi_{154}\xi_{156}+2\xi_{157}\xi_{159}+2\xi_{163}\xi_{165}+2\xi_{166}\xi_{168}+2\xi_{172}\xi_{174}\\
&+2\xi_{175}\xi_{177}+\xi_{202}\\
R_{143} = &\,\xi_{94}\xi_{204}-2\xi_{96}\xi_{136}-2\xi_{113}\xi_{161}-2\xi_{120}\xi_{170}-2\xi_{131}\xi_{204}+2\xi_{136}\\
&+2\xi_{136}\xi_{137}+2\xi_{161}+2\xi_{161}\xi_{162}+2\xi_{170}+2\xi_{170}\xi_{171}+\xi_{203}-\xi_{204}\\
R_{144} = &-2-\xi_{94}\xi_{203}+\xi_{96}^2-2\xi_{96}\xi_{135}+\xi_{113}^2-2\xi_{113}\xi_{160}+\xi_{120}^2\\
&-2\xi_{120}\xi_{169}+2\xi_{135}+2\xi_{135}\xi_{137}+2\xi_{160}+2\xi_{160}\xi_{162}+\xi_{169}^2-\xi_{170}^2\\
&+\xi_{203}+\xi_{204}\\
R_{145} = &-1-64\xi_2-128\xi_2\xi_{205}+\xi_4\\
R_{146} = &-256\xi_0\xi_2-128\xi_2+\xi_{207}\\
R_{147} = &-1-512\xi_2\xi_{207}+\xi_{206}^2-16\xi_{207}^2\\
R_{148} = &-16384-32768\xi_0-65536\xi_0\xi_{208}+\xi_2-32768\xi_{208}
\end{aligned}
$$

B Complete fixed coefficients of the ten-variable equation

This appendix defines the four integers β, K_0, M_C, M_F used in the degree-190 equation. Every index, digit, tile, finite set, sum, product and floor appearing below is part of a fixed finite recipe. None introduces an additional quantified unknown. The recipe does not inspect whether ZFC is consistent and does not search for a polynomial zero.

Provenance and notation. These are the unchanged BFS943/radix67 constants retained by source 172 at the inspected ProveIt commit `d123610bc98fef2006bff1c9466bc3f1221003e7`. The archived finite specification is `compact/source/upstream172/inherited/strict/upstream_lucas/parent/proof_sources/indexed_constants.tex`. The operative reconstruction is `compact/source/rebuild_coefficients.py`. Only the fixed-coefficient recipe is reproduced here; the historical equation surrounding that recipe is not part of the present definition. Its constant K is denoted K_0 to match the body of this paper. All other notation in this appendix is local to the finite recipe.

B.1 The four integers

Put

$$\rho = 2^{67}, \qquad B = \rho^{8807}, \qquad \beta = B - 1 = 2^{590069} - 1, \qquad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define the finite exponent set

$$
\begin{aligned}
\mathcal{E} = &\{0,\dots,942\} \cup \{983,\dots,992\} \cup \{1003,\dots,1011\}\\
&\cup \{1013,\dots,1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.
\end{aligned}
$$

Then the other three fixed integers are

$$\begin{aligned}
M_C &= B - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063},\\
M_F &= B - 1 + 4 + (2^{51}-2)(\rho^{4393}+\rho^{5495})\\
&\quad + \rho^{1013}G_{10}(\rho) + \rho^{1065}G_9(\rho) + \rho^{2195} + \rho^{3296}\\
&\quad + (2^{65}-1)\rho^{8792}G_{13}(\rho) + (2^{19}-1)\rho^{8805},\\
K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849}\\
&\quad + (\rho^{4393}+\rho^{5495})\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) &= (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),\\
\mathcal{C} &= \{(10,1023,8),(18,1013,9),(27,1003,8),(35,983,9),\\
&\quad (44,1031,1),(45,1022,1),(46,1011,1),(47,992,1),\\
&\quad (48,1095,1),(49,2195,1),(50,3296,1)\}.
\end{aligned}$$

The tiles t_i and their integer values $v(t_i)$ are specified next. The largest exponent in Φ is 3296, smaller than both 4393 and 5495. Consequently the displayed rational intermediate in K_0 expands into integer powers of ρ. The four resulting integers are positive.

B.2 The 943 indexed fixed tiles

Use the 430 actions (u_j, v_j, d_j) defined in the next subsection. Set

$$\begin{aligned}
h(q,b) &= 2(q+1)+b,\\
\mathcal{R} &= \{u_j : d_j = 1,\ 0\le j<430\}, & \mathcal{L} &= \{u_j : d_j = 0,\ 0\le j<430\},\\
r_q &= 6 + |\mathcal{R}\cap\{0,\dots,q-1\}|, & l_q &= 205 + |\mathcal{L}\cap\{0,\dots,q-1\}|.
\end{aligned}$$

An interval ending at -1 is empty. The following table defines every $t_i = (w,n,e,s)$ for $0\le i\le 942$. The index ranges and the actions are fixed data, not parameters of the Diophantine equation.

Index i and fixed range	Tile $t_i = (w,n,e,s)$
0	$(2,435,3,1)$
1	$(5,0,0,0)$
$j+2$, where $j = 2q+b < 430$	$((1-d_j)l_{u_j},\ h(q,b),\ d_j r_{u_j},\ v_j)$
$432+b$, $b\in\{0,1\}$	$(0,h(215,b),0,h(215,b))$
$434+b$, $b\in\{0,1\}$	$(0,b,0,b)$
$424+2r_q+b$, $q\in\mathcal{R}$, $b\in\{0,1\}$	$(r_q,b,0,h(q,b))$
$424+2l_q+b$, $q\in\mathcal{L}$, $b\in\{0,1\}$	$(0,b,l_q,h(q,b))$
$924+j$, $0\le j<4$	$(0,j+430\lfloor j/2\rfloor,0,438)$
928	$(2,434,2,0)$

Index i and fixed range	Tile $t_i = (w, n, e, s)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t$, $t \in \{434, 435, 436\}$	$(1, 438, 1, t)$
$934 + 3a + b - 6\lfloor a/4 \rfloor$, $a \in \{0, 1, 4\}$, $b \in \{0, 1, 2\}$	$(a, 437, b, 437)$

For each tile $t = (w, n, e, s)$, define

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127 \cdot 2^{34}.$$

B.3 A closed decoder for the 430 actions

For fixed nonnegative integers define

$$D_b(A, k) = \left\lfloor \frac{A}{b^k} \right\rfloor \bmod b, \qquad a \bmod b = a - b \left\lfloor \frac{a}{b} \right\rfloor.$$

Let $\mathcal{A}$ be the single fixed integer specified in the next subsection. First split it into three fixed blocks:

$$N = \mathcal{A} \bmod 2^{636}, \qquad T_0 = \left\lfloor \frac{\mathcal{A}}{2^{636}} \right\rfloor \bmod 2^{1675}, \qquad E_0 = \left\lfloor \frac{\mathcal{A}}{2^{2311}} \right\rfloor.$$

For $0 \le q < 215$, $b \in \{0, 1\}$ and $j = 2q + b$, put

$$\begin{aligned}
\nu_j &= D_2(N, j), & \sigma_j &= \sum_{k=0}^{j-1} \nu_k, \\
m_q &= D_2(N, 430 + q), & \tau_q &= \sum_{k=0}^{q-1} m_k, \\
F_q &= 14(1 - m_q) + m_q D_{16}(E_0, \tau_q), & & \\
u_j &= \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j), & & \\
v_j &= D_2(F_q, 2b), & d_j &= D_2(F_q, 2b + 1).
\end{aligned}$$

Empty sums are zero. In particular, $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

B.4 The 436-character numeral

The alphabet, in increasing order of digit values $0, \ldots, 61$, is

`0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following eight lines without spaces or line breaks. Interpret their 436 characters as digit values $e_0, \ldots, e_{435}$ in that alphabet, and set

$$\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}.$$

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

These data close the coefficient definitions: they determine the actions, then all 943 tiles, then Φ, and finally the four integers. The supplied reconstruction checks the exact integers against the authoritative payloads, independently of the gate evaluator. The finite decoder, tile indices, exponent ranges and numeral above agree with that reconstruction. This appendix supplies fixed coefficients; the proof of their connection to refutations is the separately identified inherited calibration.